Steady temperature distribution in an aluminium pin fin

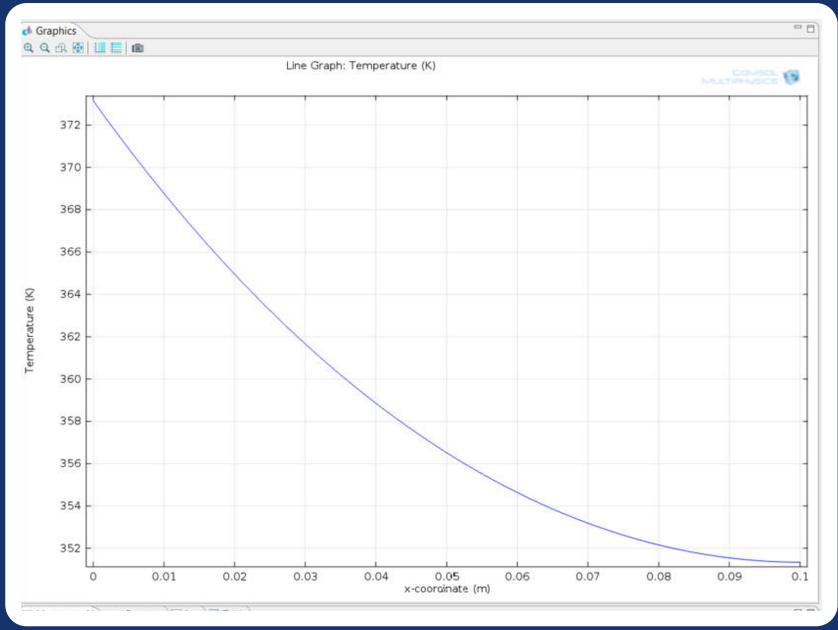
Step-by-step COMSOL Multiphysics 4.3a guide for students

Problem: 1D steady heat conduction with convection (Introduction to CFD, Problem 1)

Tool: COMSOL Multiphysics 4.3a, Heat Transfer in Solids, 1D, stationary study

Result: tip temperature 78.19 C and heat loss 2.170 W, matching the exact solution

Includes: 12 annotated steps, theory, expected results and a mesh-refinement study



Model file: fin1d_pin_fin.mph (open it in COMSOL to compare with your own model)

The problem and the idea behind the model

A cylindrical aluminium pin fin is attached to a wall at T_b = 100 C. Heat is conducted along the fin and convected from its surface to the surrounding air at T_inf = 30 C. The tip is insulated. Find the steady-state temperature distribution, the tip temperature and the heat lost by the fin.

Quantity	Symbol	Value
fin length	L	10 cm = 0.1 m
fin diameter	d	5 mm = 0.005 m
conductivity of aluminium	k	237 W/(m.K)
convection coefficient	h	25 W/(m^2.K)
air temperature	T_inf	30 C = 303.15 K
base temperature	T_b	100 C = 373.15 K
initial temperature	T_init	30 C (not needed: steady state)

Assumptions: steady state; temperature uniform over each cross-section (1D); constant k and h; radiation neglected; insulated tip.

1. Physics: energy balance on a slice dx

$$kA\frac{d^2T}{dx^2} - hP(T - T_\infty) = 0$$

$$\frac{d^2T}{dx^2} = m^2(T - T_\infty), \qquad m^2 = \frac{hP}{kA} = \frac{4h}{kd}$$

Boundary conditions: T(0) = T_b (base) and dT/dx = 0 at x = L (insulated tip).

2. Exact solution (used to check COMSOL)

$$\frac{T - T_\infty}{T_b - T_\infty} = \frac{\cosh m(L - x)}{\cosh mL}$$

$$Q = \sqrt{hPkA}\,(T_b - T_\infty)\tanh(mL)$$

With the data: m = 9.186 1/m, mL = 0.9186, T_tip = 78.194 C, Q = 2.1702 W.

3. How the problem maps to COMSOL

1D line, length 0.1 m
the fin (the cross-section is carried by the 1/A in the source term)

Temperature = 373.15 K at Point 1
base condition T(0) = T_b

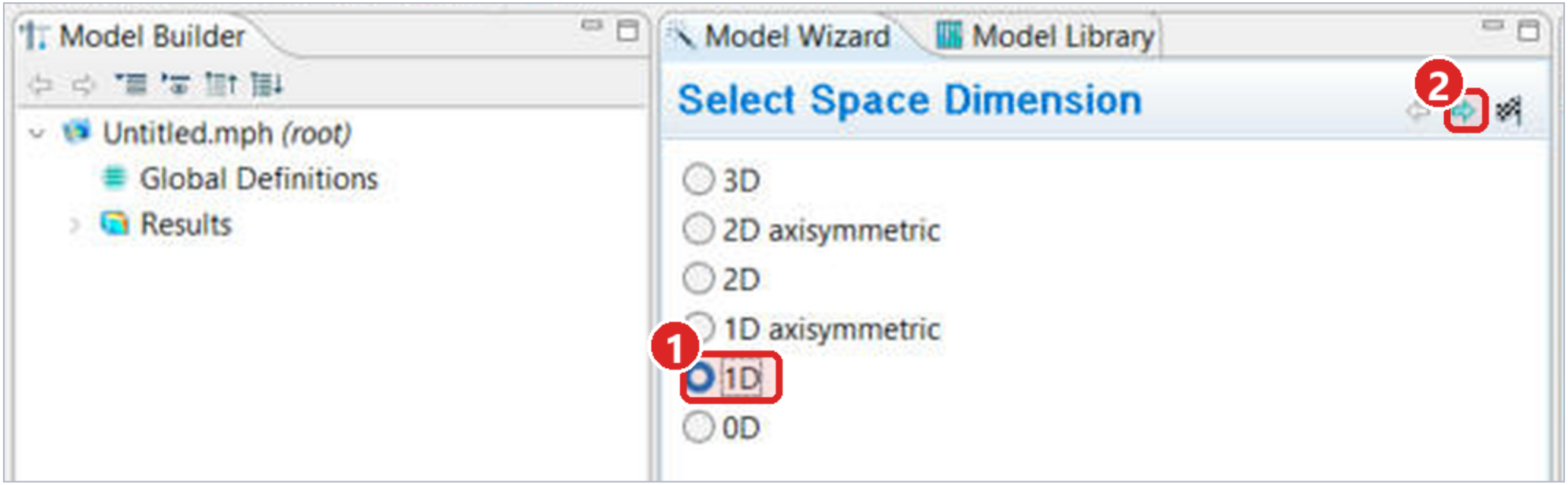
Thermal Insulation at Point 2
insulated tip (default condition, already there)

Heat Source Q = -h (4/d)(T - T_inf)
convection from the side surface, as W per m^3

Stationary study
steady state, no time stepping

Step 1 - Open COMSOL and choose the space dimension

Start COMSOL Multiphysics 4.3a from the desktop shortcut. If the Model Wizard is not showing, use File > New.

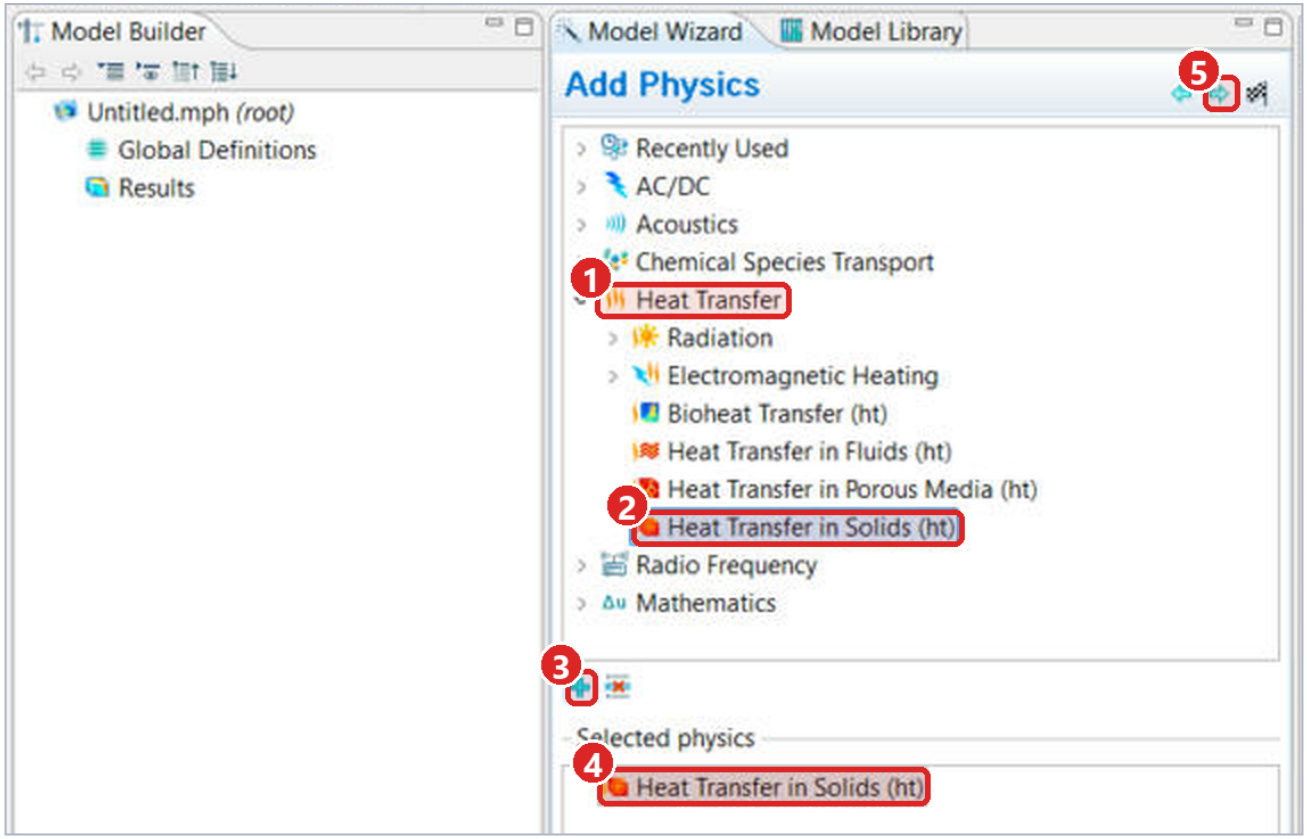


1. Select 1D. The pin is long and thin, so the temperature is assumed to vary only along the fin axis x (Assumption 2 of the problem statement).
2. Click the Next arrow at the top right of the Model Wizard panel.

Why?
A 1D model solves for T(x) only. It is the COMSOL equivalent of the one-dimensional finite-difference model in the problem: each cross-section has a single temperature.

Step 2 - Add the physics: Heat Transfer in Solids

The Add Physics page lists all physics interfaces. We need conduction inside a solid body.

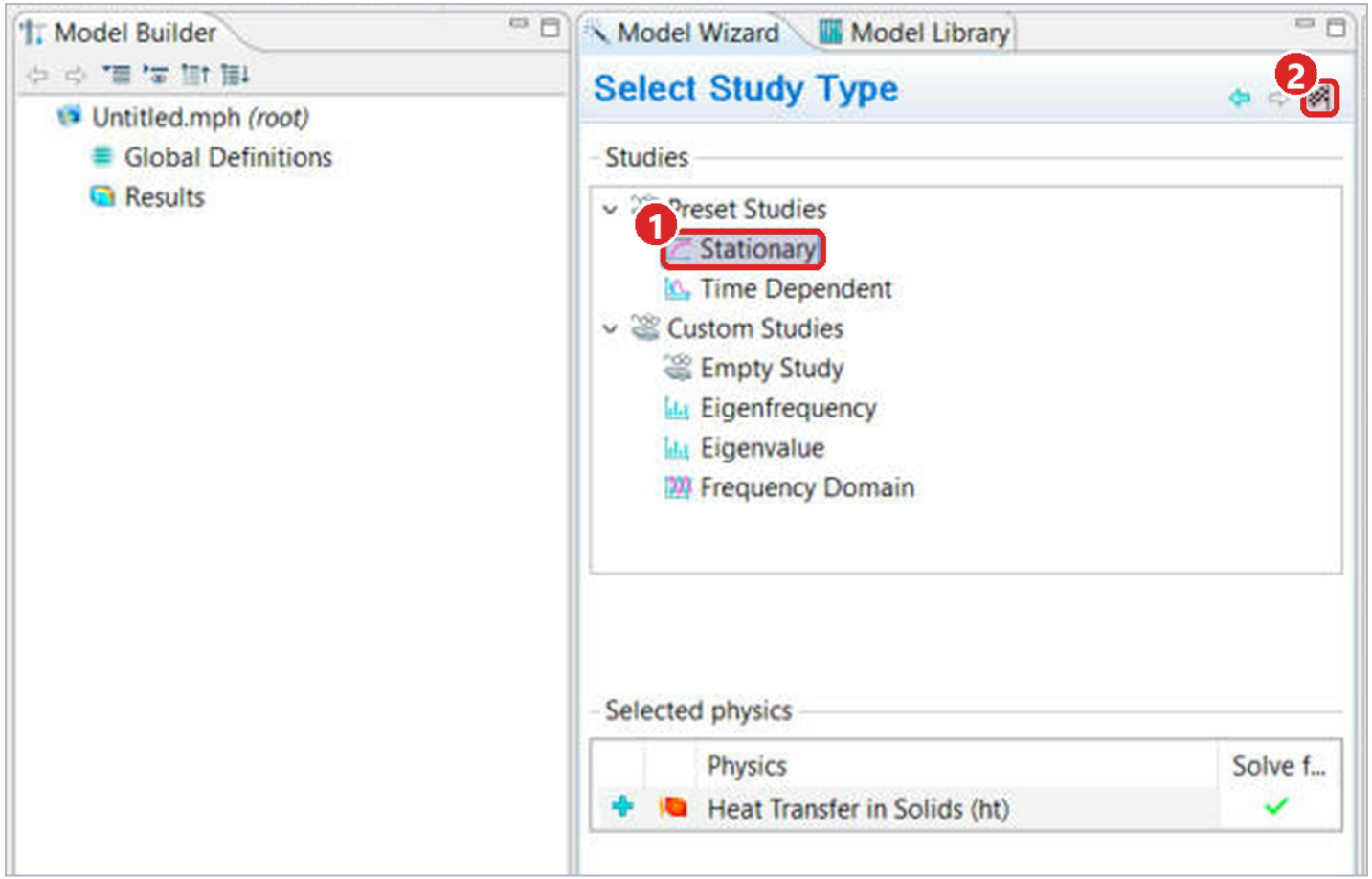


1. Expand the Heat Transfer branch (double-click it, or click the small arrow).
2. Select Heat Transfer in Solids (ht).
3. Click the blue + button (Add Selected).
4. The interface now appears under Selected physics. The dependent variable is the temperature T.
5. Click Next.

Why?
Heat Transfer in Solids solves the steady conduction equation -d/dx(k dT/dx) = Q in 1D, where Q is a volumetric heat source. We will use Q to represent the convective heat loss from the side of the fin.

Step 3 - Choose a stationary study

The last wizard page asks what kind of calculation to perform.

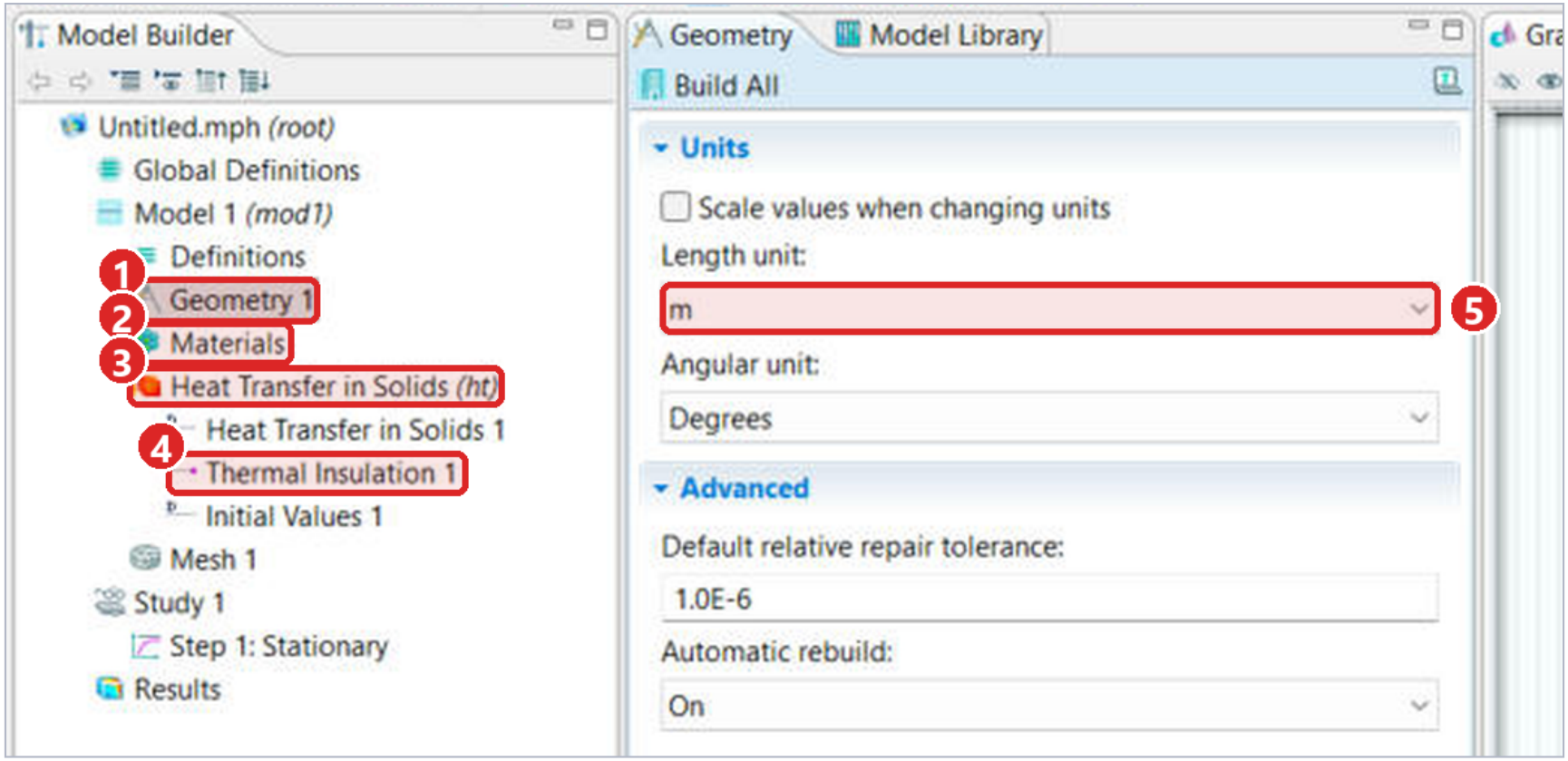


1. Select Stationary (under Preset Studies). The problem asks for the steady-state temperature distribution.
2. Click Finish (the chequered-flag icon).

Why?
The initial fin temperature of 30 C (T_init) is only needed in a time-dependent simulation. For the steady-state solution it has no influence on the final answer, so a Stationary study does not use it.

Step 4 - Tour of the model tree

After Finish the Model Builder (left panel) contains the complete outline of the model. We will work through it from top to bottom.

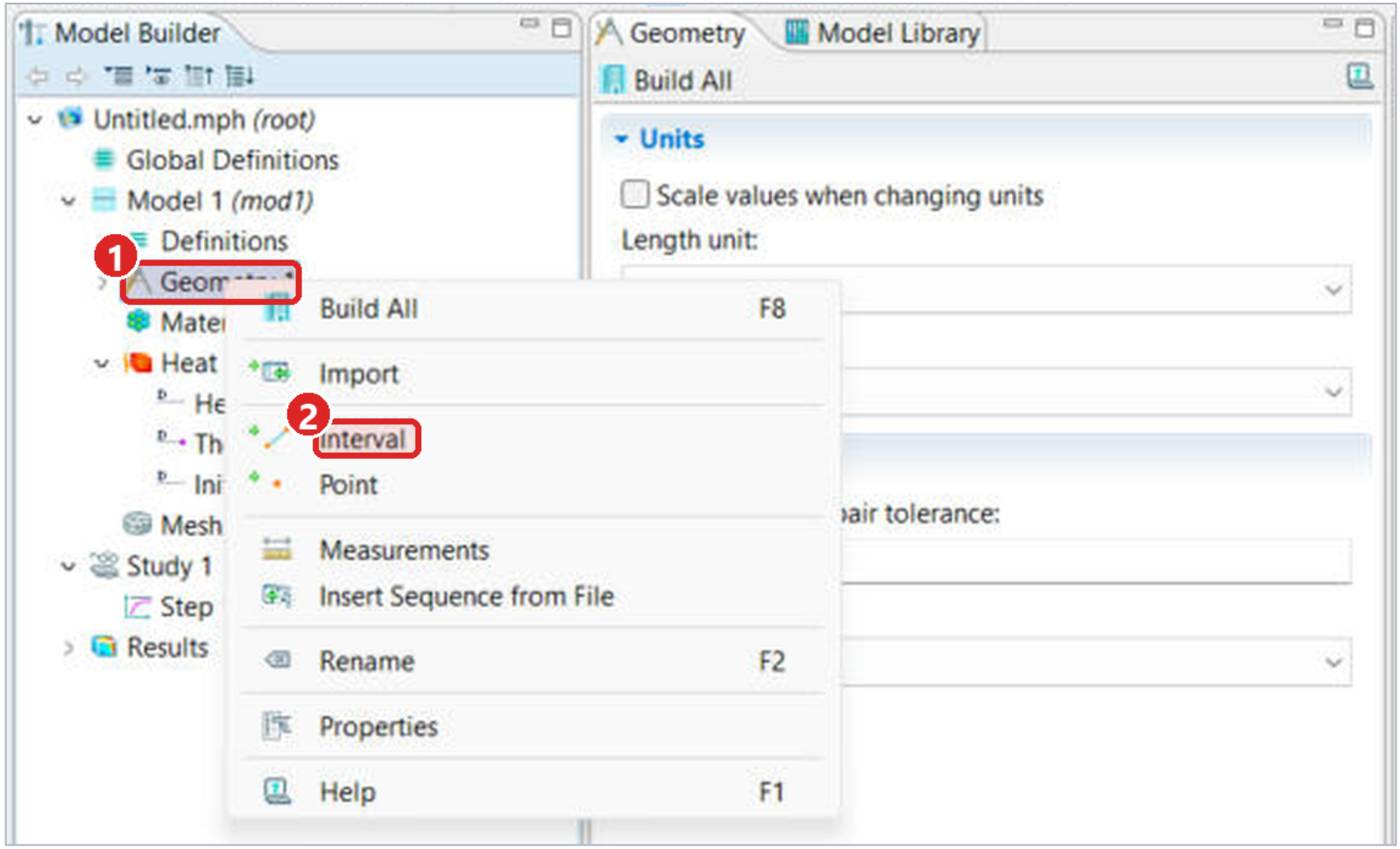


1. Geometry 1 - the shape of the fin (Step 5).
2. Materials - aluminium properties (Step 6).
3. Heat Transfer in Solids (ht) - equations and boundary conditions (Steps 7 and 8).
4. Thermal Insulation 1 is added by default on every boundary. This is exactly the insulated-tip condition of the problem, so we keep it.
5. Length unit is metre (m). We will enter all lengths in metres so everything stays in SI units.

Why?
In a 1D model the 'domain' is the line (the fin) and the 'boundaries' are the two end points: Point 1 at x = 0 (base) and Point 2 at x = L (tip).

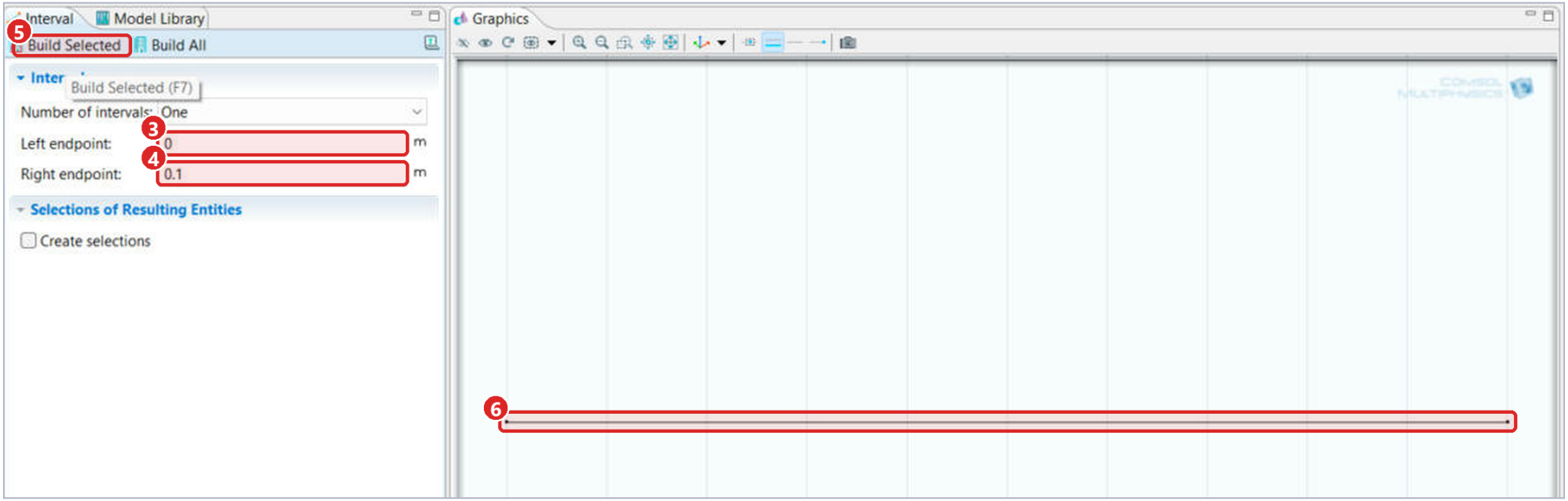
Step 5 - Build the geometry: a 10 cm line

The fin is a line of length L = 0.1 m that starts at the wall (x = 0).



1. Right-click Geometry 1.
2. Choose Interval from the menu.

Step 5 - Build the geometry: a 10 cm line

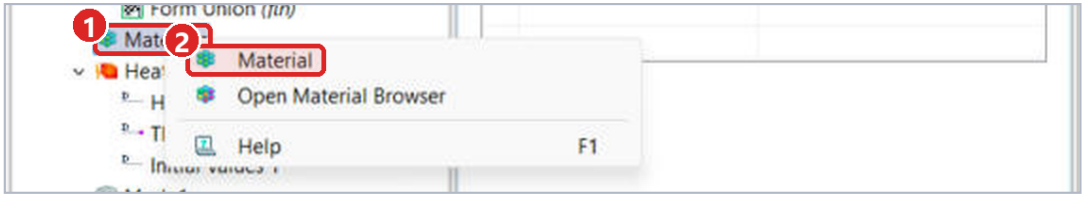


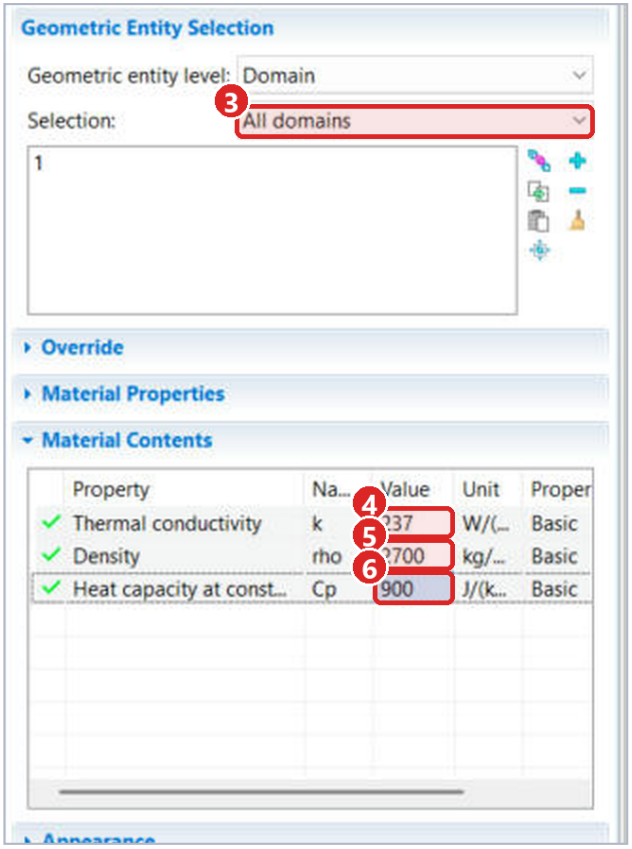
3. Left endpoint = 0 (m).
4. Right endpoint = 0.1 (m), i.e. L = 10 cm.
5. Click Build Selected (or press F8) to create the geometry.
6. The line from x = 0 to x = 0.1 appears in the Graphics window. Its end points are Point 1 (base, x = 0) and Point 2 (tip, x = 0.1).

Why?
Type 0.1, not 10: the length unit is metres. If you type a wrong number, simply correct it and build again.

Step 6 - Define the material: aluminium

We define the aluminium directly with the three properties listed in the problem data table.



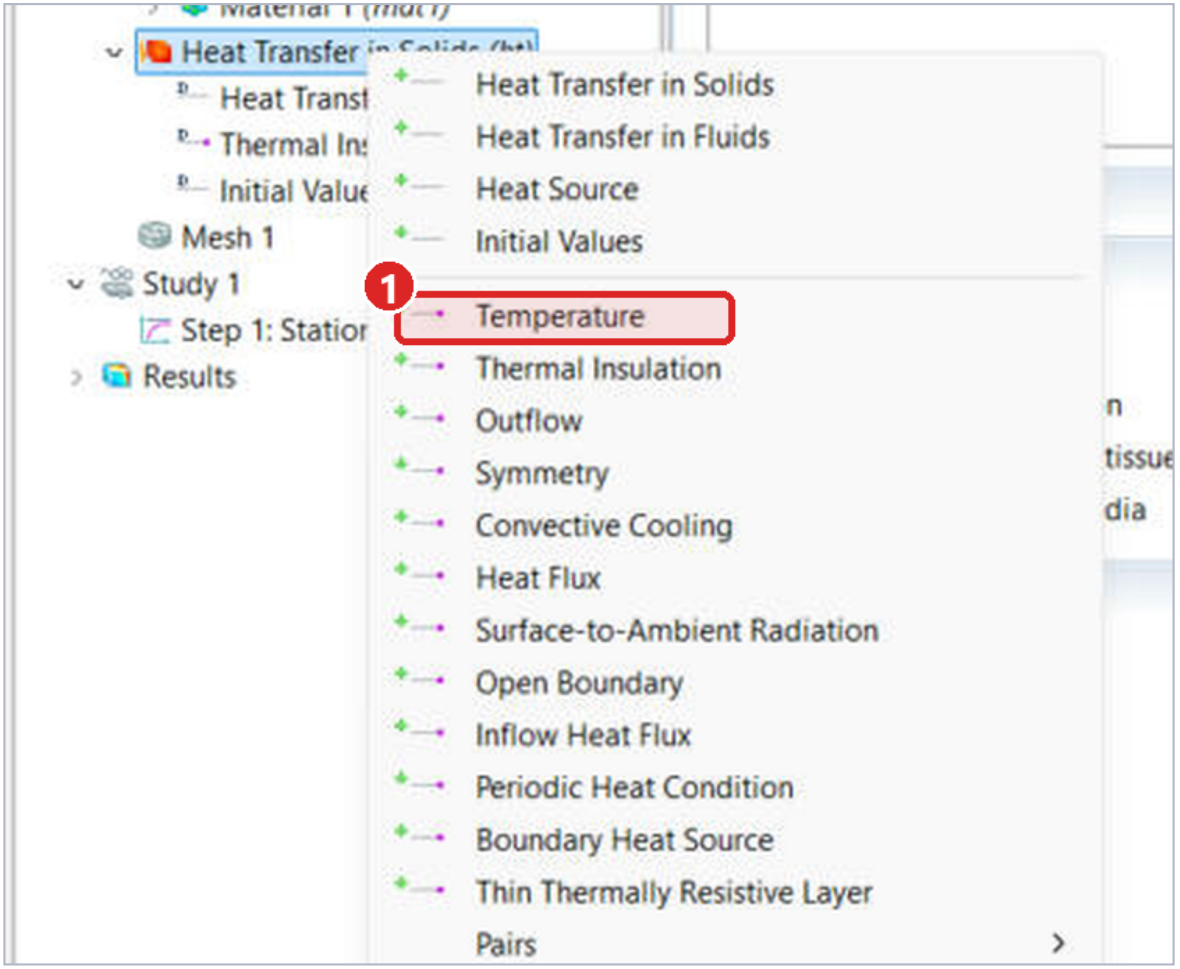


1. Right-click Materials (select it first if the menu does not appear).
2. Choose Material (a blank material).
3. Selection is All domains: the single line is made of aluminium.
4. Thermal conductivity k = 237 W/(m.K).
5. Density rho = 2700 kg/m^3.
6. Heat capacity Cp = 900 J/(kg.K). Double-click each Value cell, type the number and press Enter. The green check marks show the properties are complete.

Why?
Density and heat capacity do not affect the steady-state temperature, but COMSOL requires a complete material before it will solve. Only k matters for this problem.

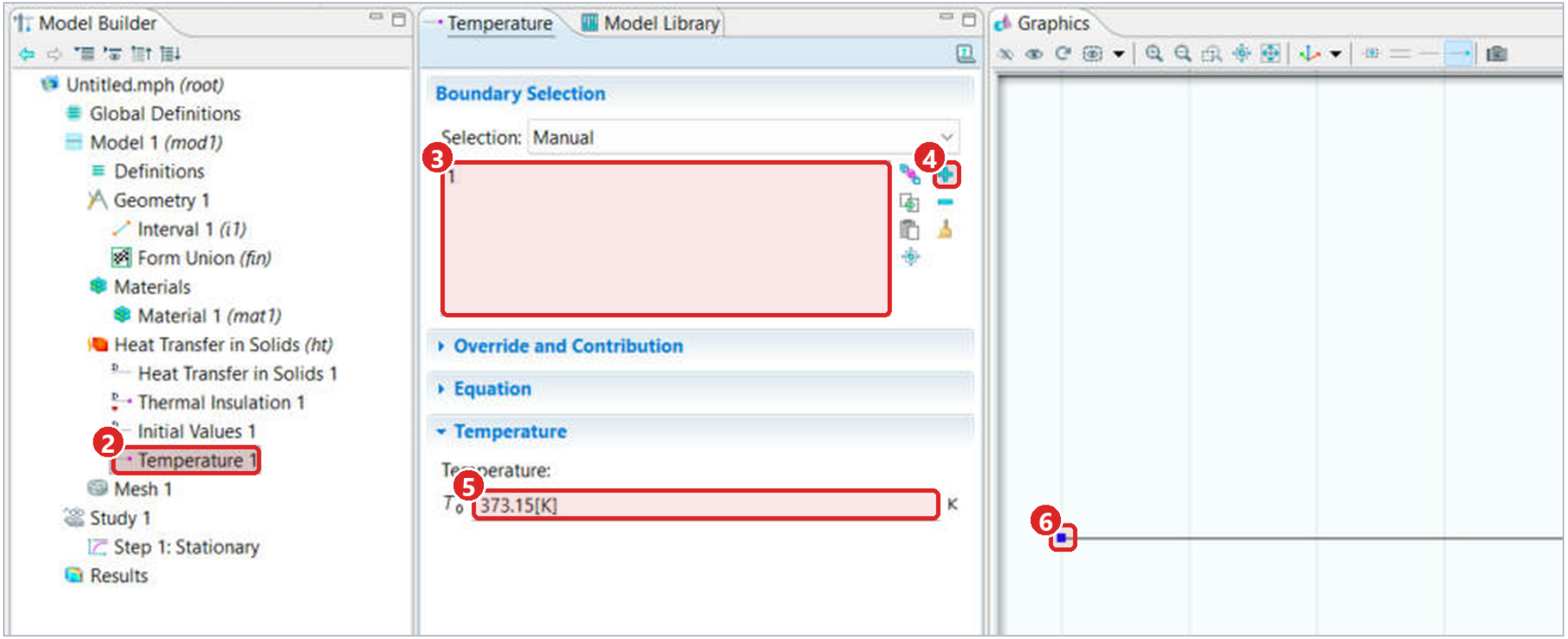
Step 7 - Fix the base temperature: 100 C

The wall holds the base of the fin at T_b = 100 C = 373.15 K. This is a Dirichlet (fixed temperature) condition at Point 1.



1 Right-click Heat Transfer in Solids (ht) and choose Temperature.

Step 7 - Fix the base temperature: 100 C

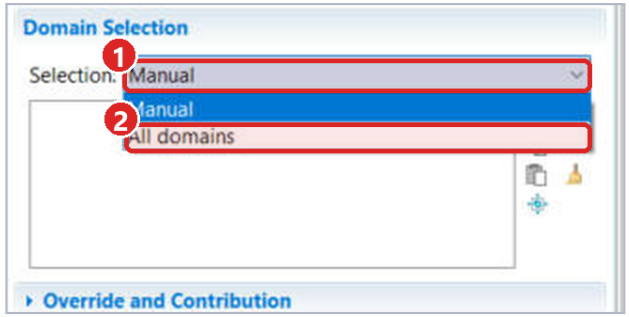


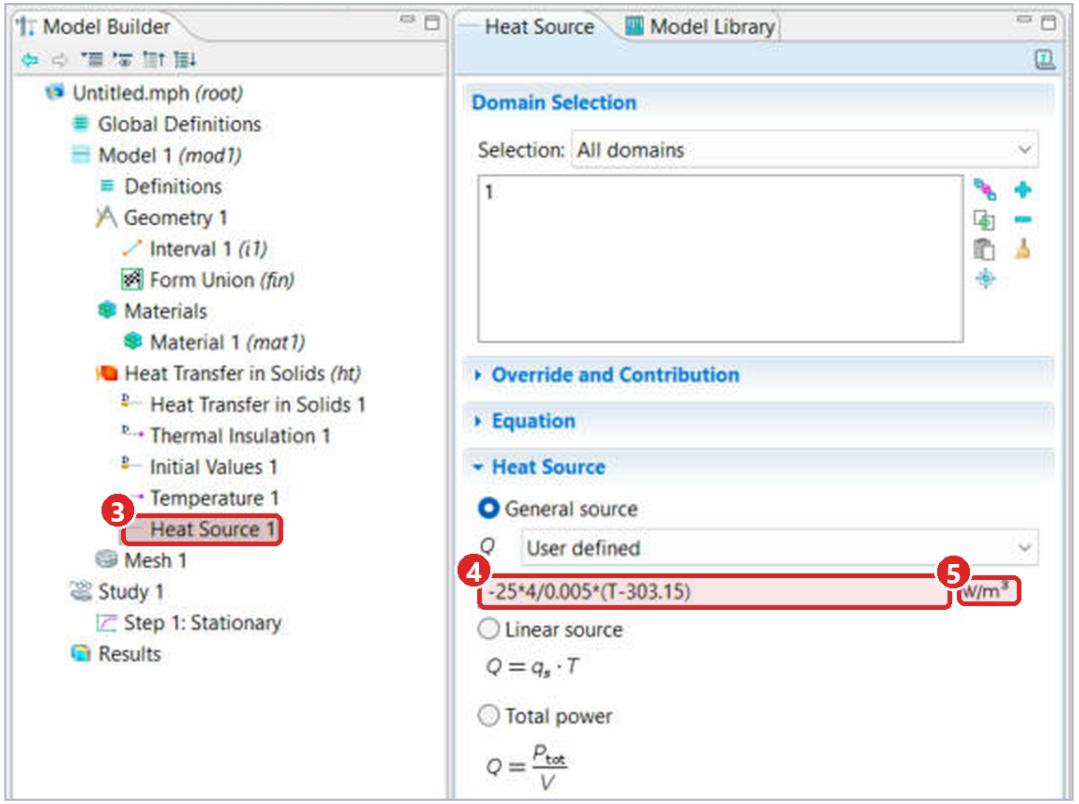
2. The new node Temperature 1 appears in the tree.
3. The selection list must contain the number 1 (Point 1, the base).
4. How to fill it: move the mouse over the left end point in Graphics until it turns red, click it, then press the + button (Add to Selection).
5. Temperature T0 = 373.15[K]. Typing the unit in square brackets keeps the entry unambiguous.
6. The selected base point is drawn in blue.

Why?
COMSOL works internally in kelvin: 100 C = 373.15 K. The default Thermal Insulation condition remains on Point 2, so the tip is insulated. The Temperature condition overrides it only at Point 1.

Step 8 - Add the convection as a heat sink

A 1D fin model has no side surface, so the convective loss from the lateral surface is added as a volumetric heat sink along the fin.





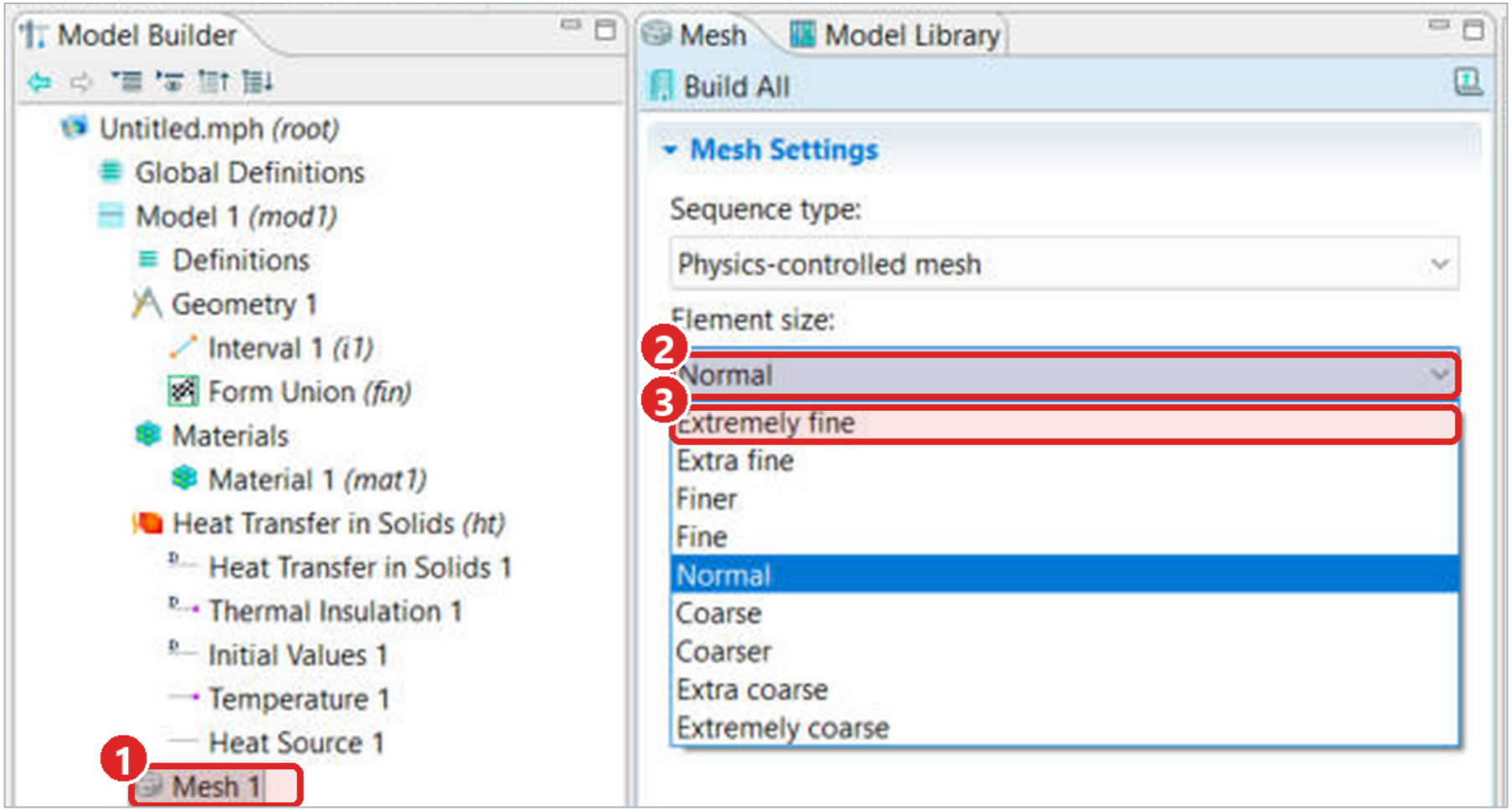
1. Right-click Heat Transfer in Solids and choose Heat Source. In its Domain Selection open the Selection list.
2. Choose All domains.
3. The node Heat Source 1 in the tree.
4. General source Q: -25*4/0.005*(T-303.15) i.e. Q = -h (4/d)(T - T_inf).
5. The unit is W/m^3 (power per unit volume).

Why?

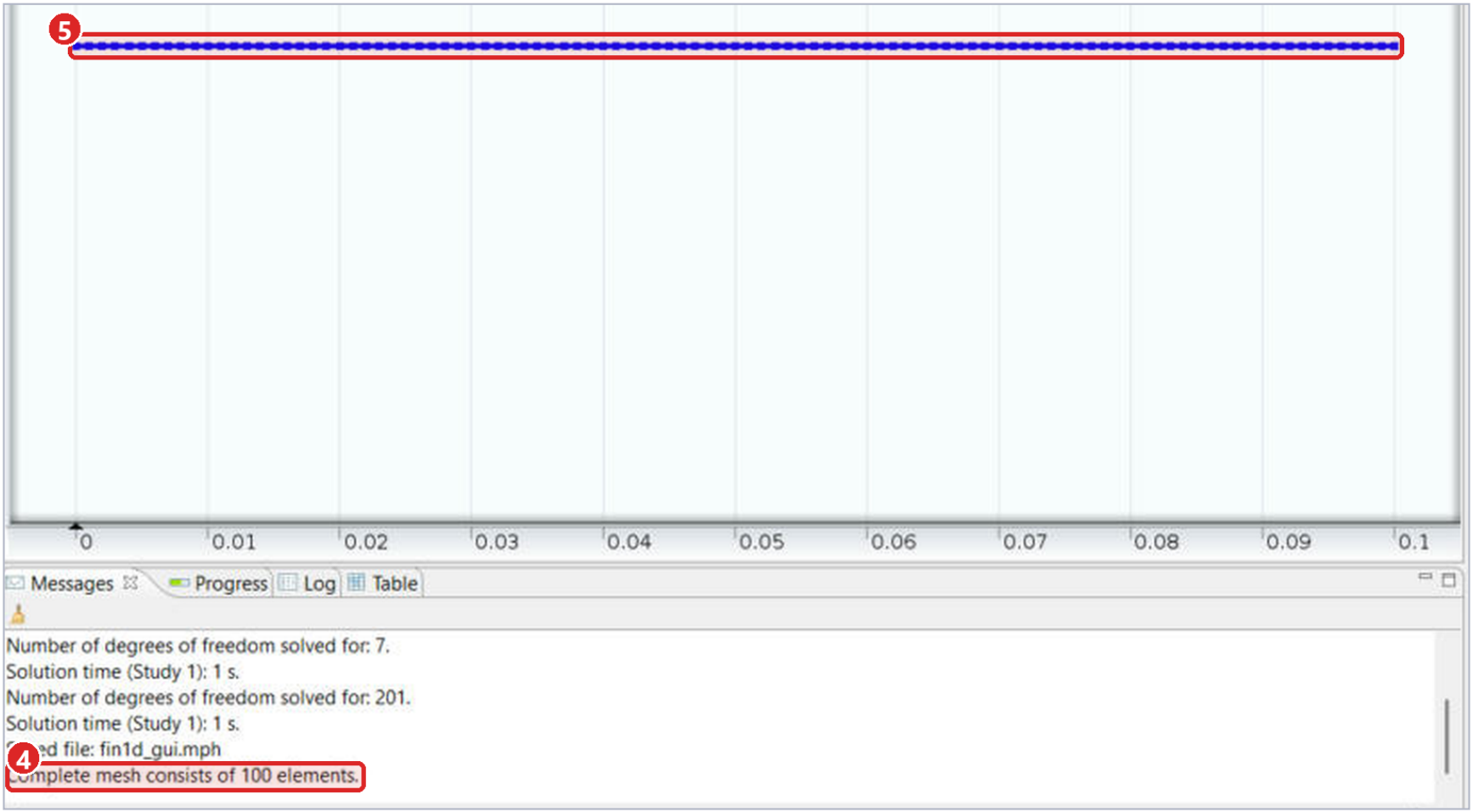
Energy balance on a slice dx: conduction in minus conduction out equals convective loss h P dx (T - T_inf), with perimeter P = pi d and area A = pi d^2/4. Dividing by A dx gives the loss per unit volume h P/A (T - T_inf) = h (4/d)(T - T_inf). Here h = 25 W/(m^2.K), d = 5 mm = 0.005 m, T_inf = 30 C = 303.15 K. The minus sign makes it a heat sink. The tip is insulated, so no extra term is needed there.

Step 9 - Create the mesh

The finite-element mesh divides the line into small elements, just like the grid of nodes in the finite-difference method.



1. Click Mesh 1.
2. Open the Element size list (sequence type: Physics-controlled mesh).
3. Choose Extremely fine, then click Build All (F8).



4 The Messages window reports: Complete mesh consists of 100 elements.

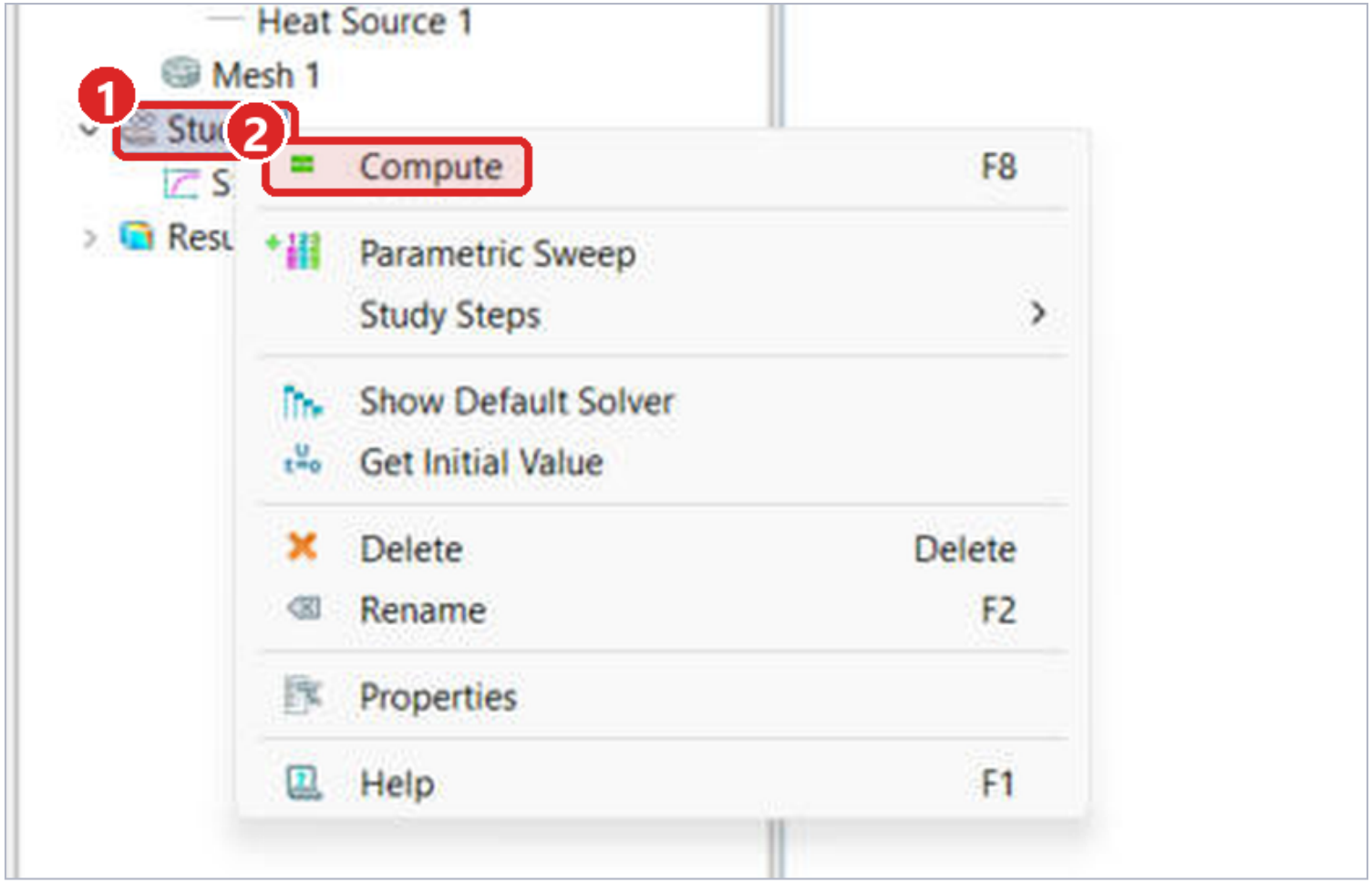
5 The mesh is drawn on the line (each dot is a mesh node).

Why?

Extremely fine gives 100 elements (dx = 1 mm) so that the result is mesh-independent. Later, try Normal (15 elements) and Extremely coarse (3 elements) to see how the answer changes (mesh study, last page).

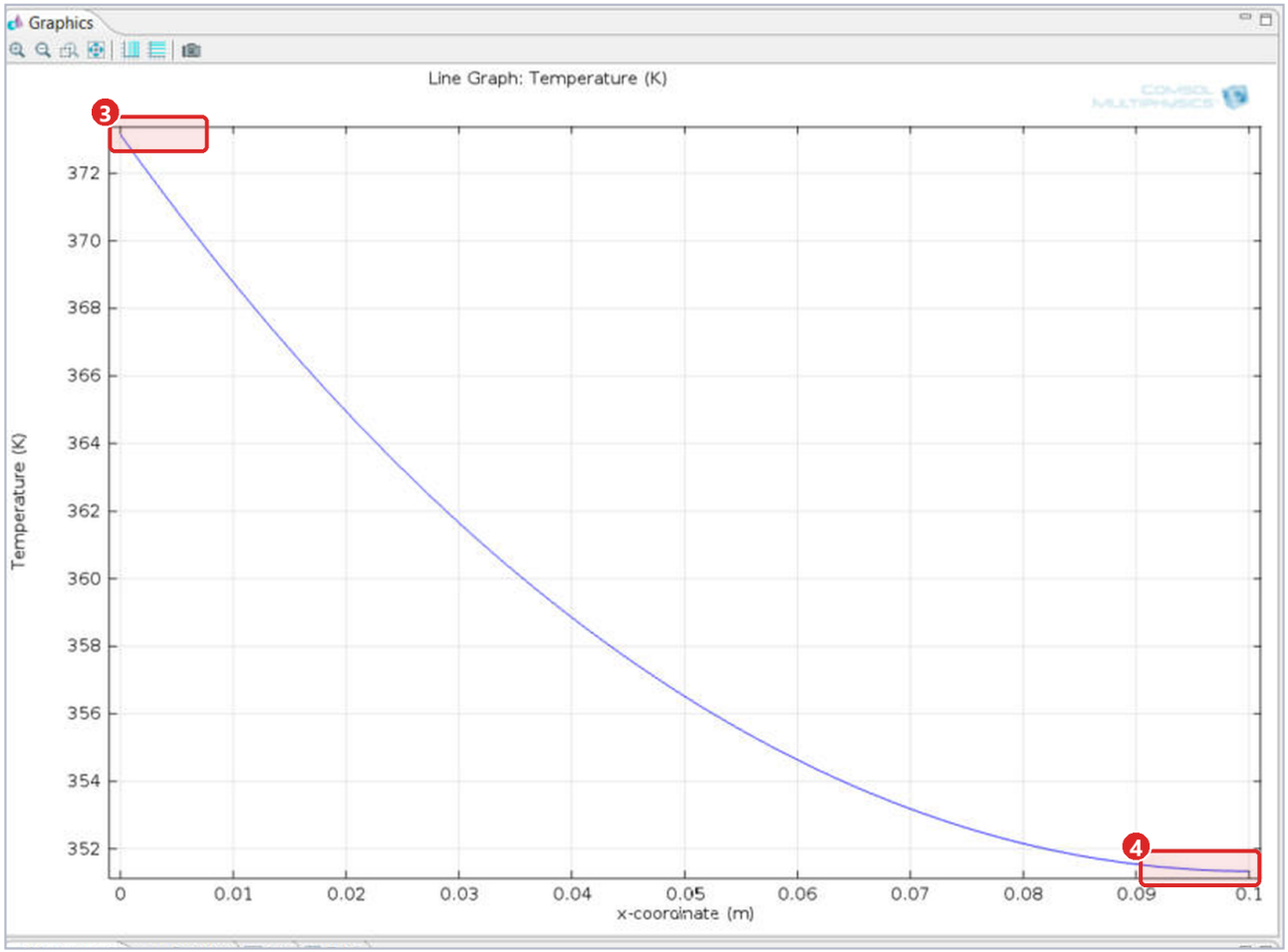
Step 10 - Run the study

Everything is defined, so we can compute the temperature distribution.



1. Right-click Study 1.
2. Choose Compute (or press F8).

Step 10 - Run the study



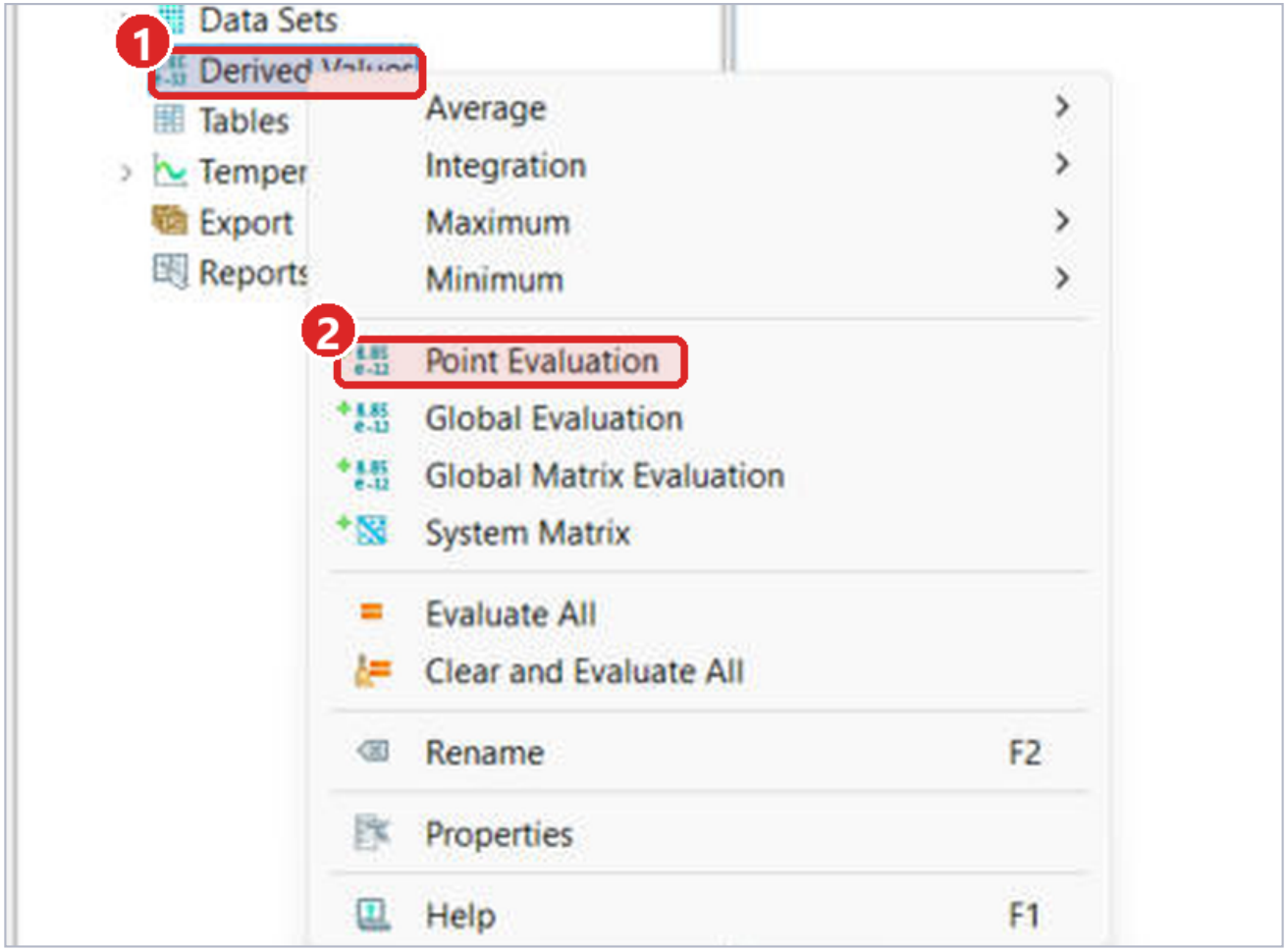
3 The default plot shows temperature versus x. At x = 0 the temperature is 373.15 K (100 C).

4 At the tip (x = 0.1 m) the curve flattens out at about 351.3 K (78.2 C): zero slope = insulated tip.

Why?
The temperature falls monotonically from base to tip. The slope is largest at the base, where the heat enters the fin, and is zero at the tip. This is the typical fin profile cosh(m(L - x)) / cosh(mL).

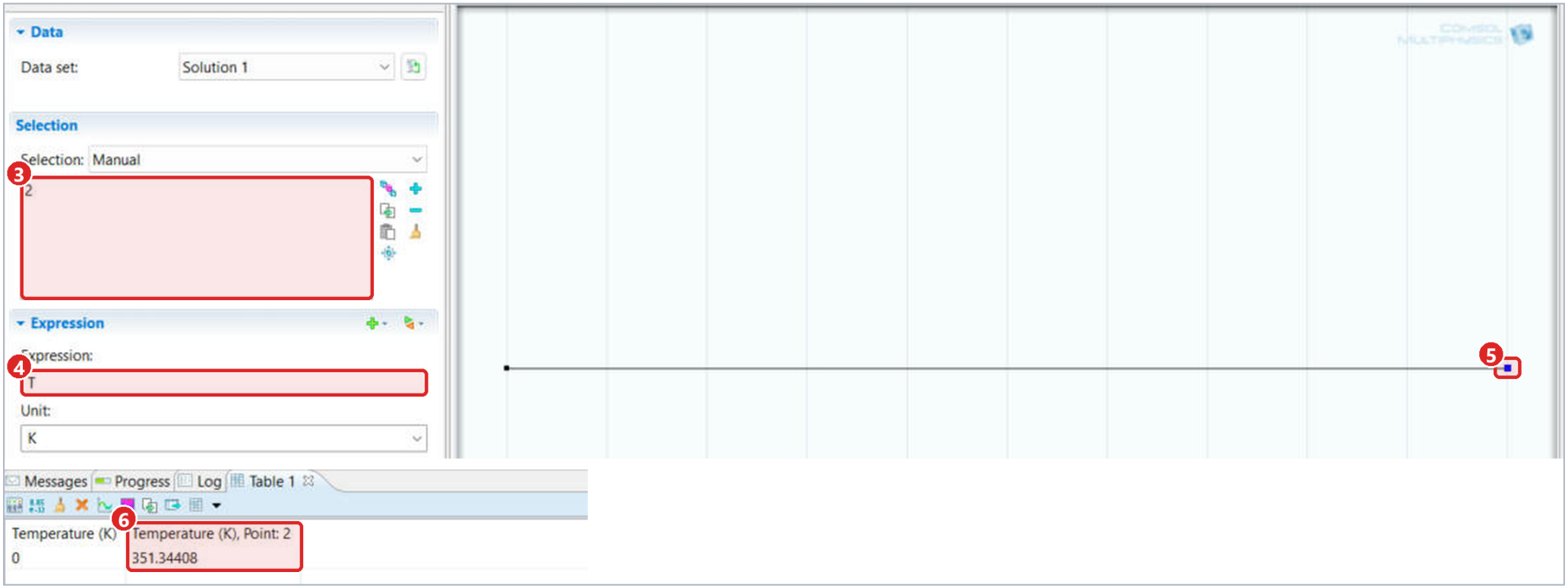
Step 11 - Read the tip temperature

To get exact numbers (not just a plot) we evaluate the solution at the tip point.



1. Right-click Derived Values.
2. Choose Point Evaluation.

Step 11 - Read the tip temperature

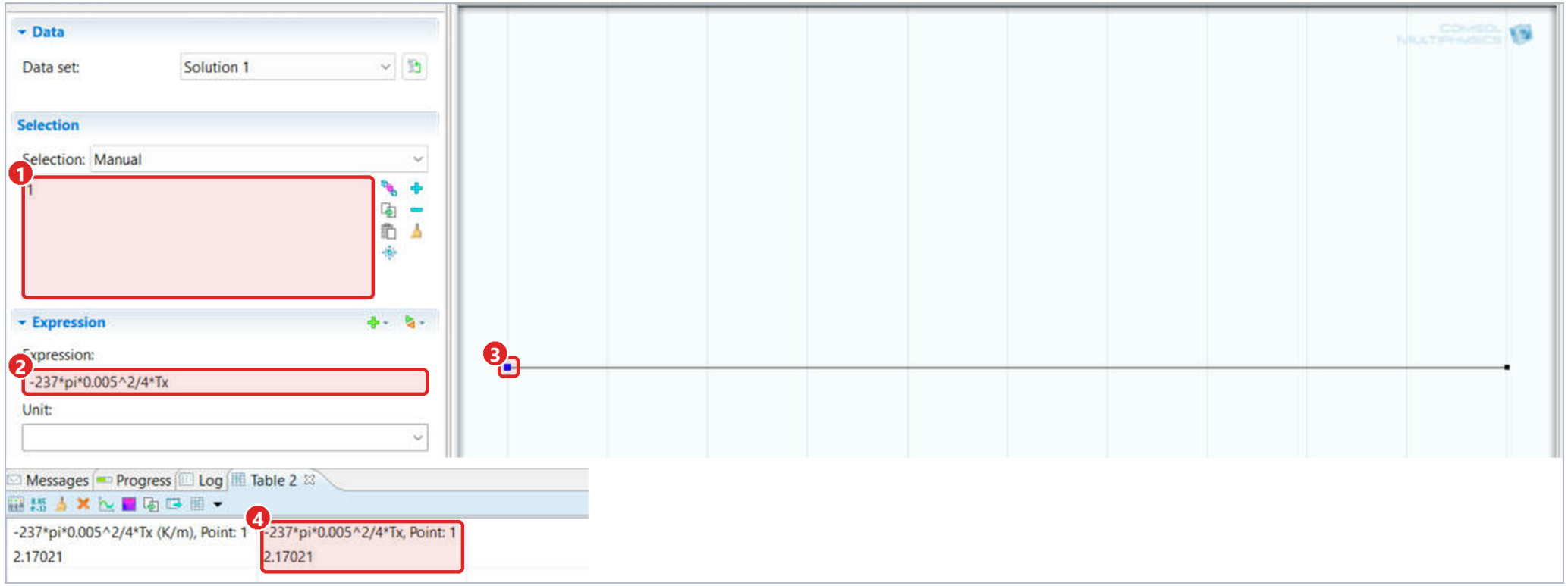


3. Select Point 2 (the tip): hover on the right end point until it turns red, click it, then press +. The list shows 2.
4. Expression T (the temperature, unit K).
5. The tip point is shown in blue. Click Evaluate (the = button at the top of the settings window).
6. The Table window shows T(tip) = 351.34408 K. (The empty first column with the value 0 comes from an earlier evaluation with an empty selection; ignore it.)

Why?
T(tip) = 351.34408 K = 78.194 C. The exact analytical solution gives 78.194 C (page 2), so the finite-element result is correct. Subtract 273.15 to convert K to C.

Step 12 - Read the heat lost by the fin

In steady state all heat that enters the fin at the base is lost by convection from its surface, so the base heat flow equals the heat lost by the fin.



1. Add a second Point Evaluation node and select Point 1 (the base): hover on the left end point until it turns red, click it, then press +.
2. Expression: -237*pi*0.005^2/4*Tx that is Q = -k A dT/dx at x = 0.
3. The base point is blue. COMSOL may fill the Unit field with K/m automatically; the expression already contains k and A, so the number is in watts. Clear the Unit field and click Evaluate.
4. Result: Q = 2.17021 W. (If a column labelled K/m also appears, it was evaluated before the unit was cleared and shows the same number.)

Why?
Fourier's law gives the heat flow into the fin: Q = -k A dT/dx evaluated at the base. A = pi d^2/4 = 1.9635e-5 m^2 and Tx is COMSOL's name for dT/dx. The minus sign turns the negative gradient into a positive heat flow. Tx at the tip is almost zero, which confirms the insulated tip.

Checking the result against the exact solution

A. Final results (Extremely fine mesh, 100 elements)

Quantity	COMSOL 4.3a	Exact solution	Difference
Tip temperature	351.34408 K = 78.19408 C	78.19408 C	< 1e-5 K
Heat lost by the fin (Q)	2.17021 W	2.17022 W	0.0005 %

B. Mesh refinement study (change Element size in Mesh 1, then Compute and Evaluate All)

Element size setting	Elements	Degrees of freedom	T_tip (K)	Q (W)	Q error
Extremely coarse	3	7	351.34427	2.15597	-0.66 %
Normal	15	31	351.34408	2.16957	-0.030 %
Extremely fine	100	201	351.34408	2.17021	-0.0005 %
Exact	**-**	**-**	**351.34408**	**2.17022**	**0**

How to read this: the tip temperature is already exact on the Normal mesh because COMSOL uses quadratic (second-order) elements. The heat flow Q needs a derivative of the solution, and its error falls roughly with the square of the element size: going from 3 to 15 elements (5 times smaller dx) reduces the error about 22 times, and from 15 to 100 elements it falls by a further factor of 40 or more (the printed digits limit how precisely this can be stated).

C. Questions for students

1. Change h to 100 W/(m^2.K) (convection with a fan). Predict first: will the tip be hotter or colder? Then check with COMSOL and the exact formula.
2. Replace aluminium by stainless steel (k = 16 W/(m.K)). How does m L change and what happens to the tip temperature?
3. Compare your COMSOL numbers with the finite-difference (Gauss-Seidel and Thomas) results from the Python notebook for N = 10, 20, 40. Which method gives the smaller error for the same number of intervals?
4. Remove the Thermal Insulation at the tip and add a Convective Heat Flux (h = 25, T_ext = 303.15 K) at Point 2. How large is the change of Q?

Troubleshooting and tips

The right-click menu does not appear.

Left-click the node once to select it, then right-click again.

A selection list stays empty after clicking in the Graphics window.

Hover the cursor over the point or line until it turns red, click it, then press the + button (Add to Selection). The number appears in the list. The minus button removes an entry.

The value shows a wrong unit or you get zero.

Check that the Selection list is not empty before pressing Evaluate; an empty selection evaluates to 0.

The results look wrong by a factor.

Check units: lengths in metres (0.1, 0.005), temperatures in kelvin for T0 and T_inf (373.15, 303.15). A temperature difference is the same in K and C.

COMSOL asks for a license.

Your computer must be on the university network or VPN so it can reach the license server. Use the license settings given in the installation guide.

Save your work.

File > Save As, choose a folder you can write to (for example Documents) and a name. Compare with the supplied file fin1d_pin_fin.mph.

Supplied files

The complete model (geometry, material, boundary condition, heat source, mesh and results) is supplied as fin1d_pin_fin.mph. Open it with File > Open to compare every node with your own model.